

Ethan Ku

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EDUCATION

University of Michigan, Ann Arbor

Graduation: May 2028

B.S.E. Electrical Engineering, Minor in Computer Science

GPA: 3.7/4.0

Coursework: VLSI Design, Monolithic Amp Circ., Semiconductor Device Phys., Princ. of ML, Data Structs. & Algor.

Languages: C/C++, Verilog-A, Verilog-AMS, SystemVerilog, MATLAB, Python, ROS2, HTML/CSS

Developer Tools: Cadence Virtuoso, Altium, LTspice, Solidworks, STM32, Git, VS Code, Linux, CATIA, Office 365

TECHNICAL EXPERIENCE

RISC Microprocessor

January 2026 – May 2026

Analog Lead, Schematic/Speedpath Lead

Cadence Virtuoso, Cadence Spectre

- Created 16-bit, 65nm RISC-architecture microprocessor, achieving a **500 Mhz** clock frequency with $53,600.88 \mu m^2$
- Virtuoso for transistor schematic and layout, Maestro for speedpath, NCVerilog for verification, and DRC and LVS
- Integrated on-chip **SRAM** caching with CPU to reduce memory-access latency between CPU and external DRAM

Stryker

January 2026 – Present

Electrical Engineering Intern

Altium, Systems Design, MATLAB, EMI/EMC, Solidworks

- Designed and manufactured **20** high-speed, impedance-matched PCBs supporting cabled and Bluetooth data conversion/transmission. Defined electrical requirements and completed schematic capture and layout in Altium
- Built **first** hardware-in-the-loop suitcase-tester robot arm for controls verification unit of **1,000+** employees
- Released **first** internal guide on exploratory Altium workflows and applied it to construct multiboard assembly of **4** cutting tool PCBs, reducing reference time to mechanical prints by **100%** by giving ownership to electrical team
- Led **2** engineering risk investigations which retired **2** risks, including 1) thermal characterization demonstrating **40%** margin to threshold, and 2) force tests that identified **1** new risk after loads were **170%** above expectations
- Redesigned existing surgical smoke evacuator to independently regulate flow to **twice** as many devices and integrated sensor technology to monitor filter health and performance, enhancing safety and efficiency in the OR

UM Solar Car Racing Team

August 2024 – Present

Microsystems Engineer, Project Lead

Altium, LTspice, Siemens NX

- Led electrical design for **4** Supermodule PCBs to unify 7 battery modules of 8-12 cells each in 4S2P hybrid configuration, eliminating legacy bus bars for safety, and maintaining **100%** functionality of high-voltage battery
- Designed a configurable Brake Light PCB with MCU, MOSFET, CAN control, and clock; consolidated multiple brake-light variants onto one board and increased constrained race-travel Brake Light replacement storage by **50%**
- Led a group of **3** to create Input Power protection boards: reverse polarity, overvoltage, and overcurrent protection
- Implemented a printf-based real-time debugger to inspect firmware execution without pausing execution sequences

Zoetics Robotics Research Group

August 2025 – Present

Research Assistant

Altium, Research Poster

- Redesigned robot compute PCB by fixing trace width violations, missing component footprints, and DRC errors
- Designed and executed publication-critical traversal test, innovating a realistic hill design with rocks for more grip
- Presented research at ECE Undergraduate Research Symposium poster session, which featured a total ≈ 20 posters

PUBLICATIONS

D. T. Islam, E. Telli, N. Telli, H. Fatteh, I. Ma, **E. Ku**, S. Kotaru, S. Hashsham, et al. (2023). “**Targeted Sequencing Panel to Characterize the Respiration of Polychlorinated Dibenzo-p-dioxins and Dibenzofurans by Dehalococcoides mccartyi Strains**”. *Annual Superfund Research Project Meeting, Albuquerque, New Mexico, 2023*

LEADERSHIP EXPERIENCE

AACCOM Chinese School

August 2022 – Present

Chinese Yoyo Teacher

- Teaches weekly Chinese Yoyo classes for up to **10** students and choreographs annual Lunar New Year performances
- Increased outreach performances by **50%**, volunteering at neighborhood performances for cultural visibility

ADDITIONAL

Extracurriculars: Taiwanese-American Student Association (Co-President), Revolution Chinese Yoyo

Spoken Languages: English (Native/Fluent), Mandarin (Native/Fluent)